

TFT_LCD_RTC_DHT.ino

```

1 #include <DHT11.h>
2 #include <TFT.h>
3 #include <SPI.h>
4 #include <Wire.h>
5 #include <RTCLib.h>
6
7 RTC_DS1307 rtc;
8 DFRobot_DHT11 DHT;
9 #define DHT11_PIN 2
10 #define cs 10
11 #define dc 9
12 #define rst 8
13
14 // create an instance of the library
15 TFT TFTscreen = TFT(cs, dc, rst);
16 String hr_str, min_str, sec_str, tmp_str, hum_str;
17 char hour_str[3], minut_str[3], second_str[3], temp[3], hum[3];
18
19 void setup()
20 {
21   TFTscreen.begin();
22   TFTscreen.background(0, 0, 0);
23   TFTscreen.setTextSize(2);
24   TFTscreen.stroke(0, 255, 0);
25   TFTscreen.text("TIME", 50, 5);
26   TFTscreen.stroke(0, 0, 255);
27   TFTscreen.line(48, 23, 108, 23);
28   TFTscreen.stroke(255, 0, 0);
29   TFTscreen.rect(5, 27, 145, 30);
30   TFTscreen.stroke(0, 255, 255);
31   TFTscreen.line(82, 64, 82, 125);
32   TFTscreen.stroke(0, 255, 0);
33   TFTscreen.text("Temp", 25, 64);
34   TFTscreen.stroke(255, 0, 255);
35   TFTscreen.text("Humi", 95, 64);
36   TFTscreen.stroke(0, 150, 200);
37   TFTscreen.text("o", 68, 82);
38   TFTscreen.setTextSize(3);
39   TFTscreen.text("C", 65, 102);
40   TFTscreen.text("%RH", 145, 95);
41   Serial.begin(9600);
42   if (! rtc.begin())
43   {
44     Serial.println("Couldn't find RTC");
45     //while (1);
46   }
47   if (! rtc.isrunning())

```

Fig. 3: Source code snippet

The DHT11 sensor is a digital smart sensor that operates on a single-wire protocol, directly providing calibrated digital outputs for both temperature and humidity. The Arduino acts as the main controller, reading data from the DHT11 sensor at intervals of about two seconds. It processes the values and updates the display with the current temperature in degrees Celsius (°C) and relative humidity in percentage (%RH).

By combining data from both the RTC and DHT11 sensors, the Arduino ensures that the TFT display presents real-time temperature, humidity, and clock information simultaneously. This integration makes the system compact, reliable, and efficient, providing a cost-effective way to monitor and display environmental conditions continuously.

Software

The code for the system is written in Arduino IDE. Before preparing the

code, it is essential to install the required libraries for the RTC module and the DHT11 sensor. Using the Library Manager in Sketch > Include Library > Manage Libraries, install the DHT11 and RTCLib libraries. A suitable display library is also required, depending on the TFT display and its driver IC.

In most cases, the display manufacturer provides a compatible library with the module. If not, popular libraries, such as TFT_eSPI, support a wide range of displays. In this design, the TFT.h library supplied with the TFT display is used. If manual installation is required, the library folder can be copied into the Arduino 'libraries' directory.

Once the required libraries are installed, the coding process begins by including and initialising all the libraries at the start of the program. This ensures communication with the RTC module, DHT11 sensor, and TFT display.

Inside the loop() function, the Arduino periodically reads the sensor data—temperature and humidity from the DHT11, and real-time clock data from the DS1307 RTC module. The collected data is processed and displayed on the TFT screen in a formatted manner. The display refreshes automatically every few seconds, ensuring that the latest time, temperature, and humidity readings are visible.

This modular, library-based coding approach makes the program easy to understand, modify, and extend for future enhancements. Fig. 3 shows a source code snippet.

Construction and testing

Download the source code, save it, and upload it to the Arduino IDE by selecting the correct board (Arduino Nano in this case) and the corresponding COM port. Once uploaded,

TABLE 2
CONNECTIONS BETWEEN TFT LCD AND ARDUINO NANO

TFT LCD Pin	Arduino Nano Pin
Vcc	5V
GND	GND
LED/BL	3.3V
CS	D10
RST	D8
DC	D9
SDA	D11
SCK	D13

TO DOWNLOAD SOURCE CODE AND RESOURCES



Scan this QR code or go to <https://sourcecode.electronicsforu.com> to access and download source code and other resources for this project

assemble the components as shown in the circuit diagram (Fig. 2) and the pin configuration in Table 2.

The design primarily uses an Arduino Nano for compactness, but an Arduino Uno can also be used since the connections remain the same. Carefully connect the DHT11 sensor, RTC module (DS1307), and TFT display to the Arduino, ensuring the power (Vcc, GND) and signal pins are correctly wired. A 10k pull-up resistor between the DHT11 data pin and Vcc, as shown in the circuit, is necessary for stable operation.

After assembly, power up the system using either a USB connection or an external 5V supply. On startup, the Arduino initialises all connected modules and begins reading real-time data from the sensors. The TFT display shows live values of room temperature (°C), humidity (%RH), and current time (HH:MM:SS) retrieved from the RTC module. **EFY**

Ashutosh M. Bhatt is M.Tech (Gold Medalist) in Embedded Systems. A lecturer in the Electronics and Radio Engineering Department, he is a GES CL-II (Gazetted Govt. Officer) officer in the Department of Technical Education, Government of Gujarat.

